Action Fidelity Without Physical Qualification: A Replicated Diagnostic Audit of Frozen PushT World Models

Anonymous Authors[1]

Abstract

Action-conditioned latent predictions are often evaluated as evidence for physical understanding and planning, but these interpretations require different measurement and decision controls. We audit two frozen RGB PushT releases using paired simulator trajectories, actual-image probes, persistence and action controls, native-goal ranking, and recorded pretraining-distribution sequences. Bounded ridge and nonlinear probes do not achieve the physical resolution needed to interpret the original latent-editing experiment, while oracle-history prediction does not clearly outperform aligned local persistence for the audited DINO release on generated trajectories. A prospectively specified replication on 128 fresh roots supports latent action-effect fidelity and specificity for both checkpoints: four contrasts have negative 99% intervals in a fixed five-member primary family, whereas a discovery-stage DINO ranking advantage is inconclusive. An eight-root numerical audit finds that encoding-layout disagreement is strongly reduced by a convolution-precision control, without establishing the cause of all historical numerical drift. These results provide a checkpoint-specific empirical case study in which replicated action fidelity coexists with unqualified physical interpretation and limited decision evidence, rather than a new general theory or demonstrated closed-loop gain.

1. Introduction

Latent world models support planning without reconstructing every pixel, but the meaning of a successful prediction depends on the evaluation. A representation may distinguish actions without recovering small physical changes; a forecast may have low latent error without selecting a useful control. Joint-embedding models and planning systems motivate these questions (Maes et al., 2026b; Zhou et al., 2025), while prior work already distinguishes physical grounding, intervention validity and decision alignment (Vakalis, 2026; Subramanian, 2026; Liu et al., 2026). We apply those distinctions to an empirical audit rather than introduce them as a new framework.

Our starting experiment asked whether physical latent edits transfer across appearance changes. That claim requires a reliable measuring instrument and an editor whose effects can be interpreted on the evaluated feature surface. Neither prerequisite was established. We therefore retain the unsuccessful qualification rather than reinterpret latent prediction accuracy as proof that the edited states are physically correct. This motivates three concrete questions: can bounded actual-image probes resolve physical state and paired changes; do predictions match executed action consequences beyond simple controls; and does the unchanged goal metric select useful candidates?

The audit evaluates the released LeWM projected-token model and an RGB DINO-WM release supplied by the Point-LeWM authors. The latter uses dummy proprioceptive input and is not a claim about every DINO-WM implementation. We first run discovery diagnostics on 128 simulator roots and 128 recorded episode windows. Follow-ups use the inspected simulator sample and are explicitly exploratory. We then specify a fixed five-contrast family and generate 128 fresh roots without changing the task distribution or selecting favorable seeds.

The independent sample supports action-effect fidelity and matched-action specificity for both checkpoints, but not the DINO forecast-over-endpoint ranking claim under its prescribed interval. Approximately half of the fresh root–branch candidate banks have no positive goal coverage, limiting decision interpretation. A separate, bounded numerical audit distinguishes encoding layout, rollout layout, recurrence-wrapper and metric-reduction changes.

Contributions and boundary. We provide (i) a preserved qualification failure with actual-image, probe-capacity and observability controls; (ii) a fixed fresh-root replication of

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Working draft, November 2026. ICML-style evidence checkpoint; not submitted or accepted.

four checkpoint-specific latent action contrasts, together with an inconclusive ranking result; and (iii) a controlled numerical audit and authenticated operational record. We do not establish absence of physical information, successful physical editing, a causal mechanism for ranking behavior, or closed-loop improvement. The novelty sought is this bounded empirical evidence, not priority for its component evaluation ideas.

2. Related Work

Predictive representations and physical grounding. LeWM learns joint-embedding dynamics from pixels, whereas DINO-WM predicts pretrained visual features (Maes et al., 2026b; Zhou et al., 2025; Oquab et al., 2023). The Point-LeWM study compares observation interfaces and supplies the specific RGB release audited here (Oberweger et al., 2026). Physics probing and physical-state grounding are established directions (Joseph et al., 2026; Yan et al., 2026). A probe's failure can reflect its capacity, observation ambiguity or distribution mismatch, not information absence. Intervention Gap and counterfactual action evaluation explicitly qualify measurement and observation bottlenecks before physical interpretation (Vakalis, 2026; Subramanian, 2026). Our negative probe findings adopt, rather than originate, that logic.

Internal interventions. Causal abstractions and temporal intervention supervision provide foundational tests of correspondences between neural and high-level variables (Geiger et al., 2021; Lippe et al., 2022). Recent world-model studies use native counterfactual references, low-rank carriers and persistent internal corrections (Liu & Chen, 2026; Yao et al., 2026); geometric editable representations address appearance-consistent semantics (Huang et al., 2026). Divergent neural interventions can invalidate simple probe interpretations (Grant et al., 2025). We report an unqualified initial editor study, not an editor that improves upon these methods.

Action effects and decision quality. Delta-JEPA and Do-JEPA explicitly target action-sensitive representations and interventional effects (Zhang et al., 2026; Resani & Shi, 2026). Our matched-action analysis is instead a diagnostic of frozen releases, with no action-effect training. Prediction–control objective mismatch is longstanding (Lambert et al., 2020); D-JEPA studies decision alignment from predictions, realized costs and outcomes (Liu et al., 2026). Comparing rankings from actual encoded endpoints and forecasts is not itself a new decision alignment method. The broader interpretability agenda likewise distinguishes decoding, dynamics, causal verification and action (Wang et al., 2026). We contribute a replicated case study within that agenda, with unsuccessful and inconclusive outcomes retained.

3. Experimental Design

3.1. Frozen Checkpoints and Interfaces

LeWM exposes a 192-dimensional projected CLS representation. The audited DINO release exposes 196 DINOv2 ViT-S/14 patch tokens, each 384-dimensional, with auxiliary channels handled by its native recurrence. We score visual features, not a decoded RGB forecast. Images are resized to 224×224 and each released normalization is applied once. DINO then natively resizes the normalized 224-pixel image to 196 pixels before its 14-pixel patch encoder, producing $14 \times 14 = 196$ tokens; the audit RGB surface and encoder input are distinct. Models remain frozen and in evaluation mode; inference is float32 on an RTX PRO 6000 Blackwell GPU with PyTorch 2.10.0+cu128. Source, weight and normalization hashes are fixed (Appendix A). Raw latent magnitudes across these spaces are not model rankings.

3.2. Paired Simulator and Recorded Data

PushT follows the released simulator lineage (Chi et al., 2023; Maes et al., 2026a; Oberweger et al., 2026). A simulator root defines a factual and counterfactual initial-condition branch, with position or orientation intervention applied before the observation history. Both branches execute the same eight action candidates and are rerendered with original, color-only and background-only appearances, without changing their physics. The 512-unit arena and angular labels provide native physical ground truth; these units are not resized-image pixels.

Three history frames occur after primitive controls 0, 5 and 10. Each action chunk packs five ordered XY controls; two chunks connect the history, followed by eight future chunks. Thus horizon 8 ends after 50 primitive controls, including the ten-control prefix. Candidate 0 is hold; the bank also contains approach, away and fixed-seed smooth random/biased controls. Trajectories are not shortened upon native success. Apart from the original initial-geometry rejection rule, no outcome-based rejection or replacement is allowed.

The original sample has 256 editor, 128 scorer, 64 validation and 128 pilot roots. Actual-image probes fit scorer roots and select only on validation roots. All seeds and pilot candidates are retained. The recorded diagnostics use 128 authenticated episode/start windows from the released PushT expert-training dataset, identical for both models. Finite terminal action labels and finite nonterminal controls outside the native unit box are preserved without clipping; terminal rows never become outgoing transitions. This is a *pretraining-distribution diagnostic*, not held-out generalization.

3.3. Discovery, Replication and Experimental Unit

Discovery diagnostics and follow-ups use the same 128 inspected pilot roots. Their 5,000-draw intervals are pointwise 95% root/episode bootstrap intervals, not simultaneous confirmation. The follow-up protocol was fixed after earlier results but before its execution. It is not retrospective preregistration.

The subsequent replication fixes pilot seed 2026110201 and root indices 0–127 before generation. Identities are disjoint from every original split, the originally reserved 3,384-root roster and a software-preflight root. Original scorer/validation files are copied byte-identically but are not fitted or encoded in this dynamics-only pass. The unchanged generator produces exactly 128 fresh roots. Both models evaluate every branch, candidate, appearance and future step, yielding 6,144 bank/horizon records per model.

The primary slice is original appearance, horizon 8. Each bank first averages its candidate contrasts, then the two branches are averaged within a root, and roots receive equal weight. Candidates, branches, frames and appearances are not independent experimental units. The five primary contrasts share 10,000 PCG64 bootstrap draws with seed 20261102. Two-sided percentile 99% intervals apply Bonferroni correction to a nominal 0.05 familywise level; bootstrap coverage is asymptotic, not a finite-sample guarantee. A prescribed negative direction is supported only if the whole interval is below zero. All other slices and metrics remain pointwise exploratory 95% results.

4. Diagnostic Estimands and Controls

4.1. Physical Measurement

Actual-image probes predict pusher XY, block XY and sine/cosine angle labels. Ridge candidates use fixed spatial pooling and regularization grids; a 64-unit SiLU MLP uses seeds 11, 29 and 47, two weight decays and checkpoints at 50, 100 and 200 epochs. MLP feature normalization and target means are estimated only from scorer data; target scales are fixed, not estimated. Target-group selections use validation losses with separate fixed reporting scales (Appendix B). Choices and weights are published before pilot evaluation. All MLP seeds are reported, not a best seed. The follow-up includes all candidates, unlike the earlier selected-candidate mixed-time readout summary (Appendix B).

State distances and circular angular errors are paired with effect-error controls. A training-mean readout implies zero change between factual and counterfactual branches. The 4-unit/2.5-degree follow-up resolution targets are diagnostic commitments motivated after discovery, not externally established task tolerances. We do not reinterpret a relative improvement over a weak baseline as adequate absolute resolution.

4.2. Prediction and Feedback

Let $A_{rbk,h}$ be an actual encoded future for root r, branch b, candidate k, horizon h, and $P_{rbk,h}$ its autonomous forecast. Within a model, use $d(x, y) = D^{-1}\|x - y\|^2$ over every audited visual coordinate. Autonomous prediction is compared with persistence of the final initial-history feature, and with a native zero-future-action rollout preserving the prefix.

Oracle-history predictions receive actual aligned observations at each step. Their control is *local persistence*, the immediately preceding actual feature, not persistence of the initial frame. Otherwise oracle improvement would conflate useful local dynamics with a much easier prediction problem. DINO crosses free/actual visual history with recurrent/encoded-zero auxiliary history. Reset means the native encoder applied to zero proprioception, not literal zero channels; recurrent patch-varying outputs are retained. Oracle observations are privileged and not a deployable forecasting improvement.

4.3. Action-Effect Fidelity and Specificity

For $k > 0$, define hold-relative effects $\Delta P_k = P_k - P_0$ and $\Delta A_k = A_k - A_0$ within a bank. The fidelity contrast is

$$F_{rb} = \frac{1}{K-1} \sum_{k=1}^{K-1} \left[d(\Delta P_k, \Delta A_k) - d(0, \Delta A_k) \right]. \quad (1)$$

Negative favors the forecast over predicting no effect. Zero-effect cases remain without denominator floors or exclusions. Specificity compares each prediction with its corresponding actual endpoint and *every* wrong assignment:

$$S_{rb} = \frac{1}{K} \sum_{k=0}^{K-1} \left[d(P_k, A_k) - \frac{1}{K-1} \sum_{j \neq k} d(P_k, A_j) \right]. \quad (2)$$

The complete $K \times K$ assignment matrix is retained; a selected shuffle is not the control. Aggregate signed effect gain is the ratio of equal-root mean dot product to mean true-effect energy, not the mean of unstable rootwise ratios. Undefined bootstrap denominators are reported. These metrics establish latent correspondence, not block-specific physical correctness.

4.4. Finite-Bank Decisions

For each bank, apply the checkpoint's unchanged native goal score to either actual encoded endpoints or forecasts. LeWM uses summed squared feature error and DINO uses mean squared feature error; multiplying a within-model score by

Table 1. Discovery actual-image follow-up, original appearance/horizon 8. Block errors are native distance units; angles are degrees. All MLP seeds are retained. Paired angular-effect error is worse than zero change for every fitted method. Probe failures do not prove absence of physical information.

Model	Probe	Block	Angle	Effect angle
LeWM	Ridge	29.32	20.05	14.69
LeWM	MLP 11	40.67	18.92	13.98
LeWM	MLP 29	39.64	18.74	14.02
LeWM	MLP 47	38.57	19.22	14.60
DINO (Point)	Ridge	32.16	34.83	22.65
DINO (Point)	MLP 11	56.90	38.61	24.99
DINO (Point)	MLP 29	60.29	37.37	24.90
DINO (Point)	MLP 47	55.27	37.82	23.77
Zero-change	–	–	–	11.73

a positive constant does not change its ranking. Ties select the smallest index. Evaluate the chosen candidate using the saved native coverage cost $c_k \in [0, 1]$:

$$R(q) = c_{\arg\min_k q_k} - \min_k c_k. \tag{3}$$

Hold regret is $c_0 - \min_k c_k$ and expected uniform regret is $K^{-1} \sum_k c_k - \min_k c_k$. We retain absolute best coverage and the rate at which every candidate has zero coverage. Zero regret on such a bank is uninformative about success. Forecast-minus-endpoint regret is primary only for DINO; both action contrasts are primary for each model. No closed-loop task success is asserted.

5. Results

5.1. The Physical-Editor Interpretation Remains Unqualified

The original editor pilot used a scorer fitted on mixed actual/predicted surfaces. Original-appearance losses were 71.6784 for LeWM and 520.6246 for DINO, against a draft scaled-loss tolerance of 0.05. Selected MLP-minus-identity effect losses were positive (+1.0984/+7.7819) with intervals above zero. Transport interaction intervals crossed zero. These scores under *unqualified measurement* neither demonstrate physically harmful edits nor establish appearance-invariant editing. They are not directly comparable with the later actual-image-only probe experiment.

The bounded actual-image follow-up does not rescue physical resolution (Table 1). Ridge block errors are approximately 29–32 units; all six MLP-minus-ridge block-error intervals are positive. Every fitted method has greater paired angular-effect error than zero angular change. Relative improvements in other quantities therefore do not qualify small physical edits. The result is bounded to this probe family and data surface, not an impossibility theorem.

Observation limitations are substantive: 280 exact resized-RGB hash groups have different state labels, covering 630 of 6,912 original-appearance samples. The empirical single-image scaled-MSE lower bound is 3.6240. This is a bound for the finite labeled observation sample and that scaled loss, not a population, angular-distance or full-history bound. At horizon 8, 650/2,048 pusher circles are wholly outside the canvas. Fully-in-canvas ridge pusher errors still average 87.20/70.93 units (LeWM/DINO). No samples are removed from primary results.

Table 2. Actual-history prediction minus aligned local persistence at horizon 8. Negative favors prediction; intervals are pointwise exploratory 95%. Recorded sequences overlap the pretraining distribution.

Sample	Model	MSE contrast [95% CI]
Discovery	LeWM	-0.0094 [-0.0135, -0.0057]
Discovery	DINO (Point)	-0.0028 [-0.0131, 0.0072]
Recorded	LeWM	-0.2006 [-0.2392, -0.1649]
Recorded	DINO (Point)	-0.2185 [-0.2428, -0.1954]
Fresh roots	LeWM	-0.0112 [-0.0148, -0.0078]
Fresh roots	DINO (Point)	0.0029 [-0.0070, 0.0130]

5.2. Oracle Feedback Requires an Aligned Baseline

In discovery generated trajectories, replacing autonomous feedback with actual history reduces MSE from 0.3364 to 0.0279 for LeWM and from 0.7900 to 0.1555 for DINO. Those large reductions alone do not establish useful local prediction. Table 2 compares oracle forecasts with local persistence. LeWM has a negative contrast; DINO's generated contrast remains inconclusive in both discovery and the fresh sample. The fresh-sample view is exploratory, not an additional primary-family claim.

Both models have clear local advantages on recorded training-distribution windows, but this does not establish generated-task or unseen-distribution generalization. DINO auxiliary reset changes are tiny (discovery autonomous reset-minus-recurrent MSE about 2.05×10^{-5}), and do not rescue its generated local-persistence contrast. We preserve the full visual/auxiliary control rather than attribute the gap solely to autoregressive error.

5.3. Latent Action Fidelity Replicates

Discovery effect-error-minus-zero intervals were negative for both models, as were matched-minus-all-wrong assignment intervals. The fresh-root experiment supports all four action contrasts with the prespecified 99% intervals (Table 3; Figure 1). Neither model is trained, selected or refitted on the fresh roots.

Fresh-sample signed gains are 0.7649 for LeWM and 0.6644 for DINO, with pointwise 95% intervals below one and no undefined bootstrap denominators. These ratios do not im-

Table 3. All five primary fresh-root contrasts, original appearance/horizon 8. Equal-root estimates and two-sided 99% bootstrap intervals form the fixed five-member family. Negative directions support the corresponding contrast. The DINO ranking result is inconclusive, not a successful replication.

Model	Contrast	Estimate [99% CI]	Prespecified decision
LeWM	Effect vs. zero	-0.3753 [-0.4529, -0.3035]	Supported
LeWM	Matched vs. wrong	-0.5180 [-0.5842, -0.4544]	Supported
DINO (Point)	Effect vs. zero	-0.2364 [-0.2889, -0.1834]	Supported
DINO (Point)	Matched vs. wrong	-0.5605 [-0.5968, -0.5245]	Supported
DINO (Point)	Forecast vs. endpoint	-0.0071 [-0.0189, 0.0051]	**Inconclusive**

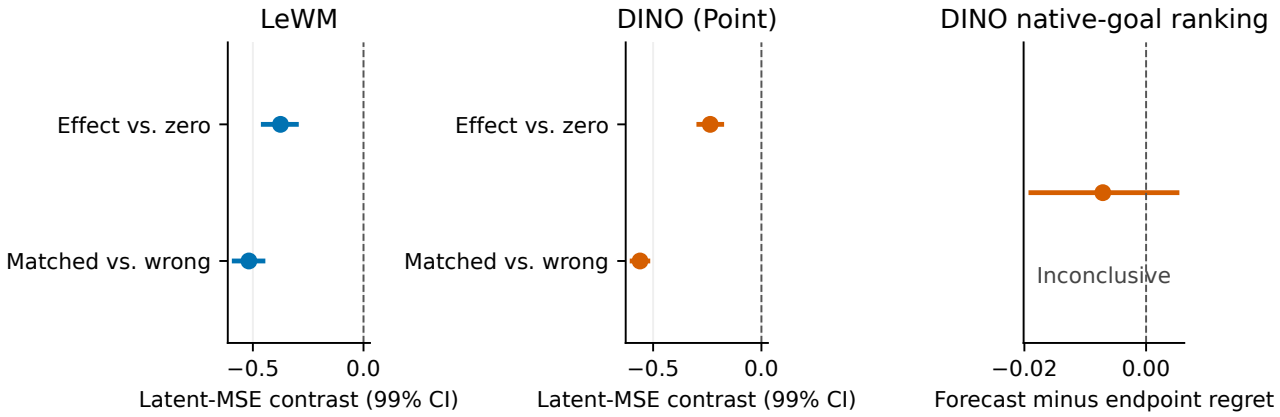


Figure 1. Fresh-root primary results. Both action contrasts are supported for each model; DINO forecast-minus-endpoint regret spans zero. Separate latent axes are intentional: raw magnitudes do not rank the models. All 128 roots, both branches and all candidates contribute; no outcome-based filtering.

ply simply insufficient amplitude: discovery predicted/ true effect-energy ratios were approximately 1.017/1.089. Directional and off-axis errors can coexist with similar total energy. More importantly, neither a negative fidelity contrast nor a positive signed gain identifies which physical component is accurately represented.

5.4. Ranking Evidence Is Limited by Both Uncertainty and Coverage

The discovery DINO forecast-minus-endpoint regret was -0.01356 with a pointwise 95% interval below zero. The fresh estimate is -0.00711, but its primary 99% interval spans zero. Therefore the discovery observation is not promoted to independently supported decision evidence. This change is not proof that the underlying effect is exactly zero; both sample variation and the stronger prespecified interval matter. We do not extend the sample or search for a seed that reproduces the discovery interval.

Expected uniform regret is 0.05083 and hold regret is 0.02701 on the fresh banks. Forecast rankings show advantages relative to expected uniform choice, but neither establishes a forecast-over-hold gain under the paired interval. All 129/256 zero-coverage banks remain, compared with 127/256 in discovery. Their weak candidate support limits discrimination among decision rules. A redesigned reachable/contact-rich bank would change the estimand and needs a new prospective study, not retrospective removal of misses.

Table 4. Fresh-root finite-bank rankings, original appearance/horizon 8. Regrets are native coverage-cost units. The paired forecast-minus-hold contrast has a pointwise exploratory 95% interval; neither interval is wholly negative.

Model	Endpoint	Forecast	Forecast–hold [95%]
LeWM	0.0201	0.0304	0.0034 [-0.0047, 0.0116]
DINO (Point)	0.0308	0.0237	-0.0033 [-0.0127, 0.0046]

5.5. Execution Layout Has a Measurable Numerical Effect

The old diagnostics encoded context in batches of four then two frames and rolled out one candidate at a time. The follow-up encoded all six context frames together, future frames in batches of 32, and eight-candidate banks. Historical individual native errors differed by up to 0.028922/0.013884; aggregate native/persistence changes were at most 0.000146. We retain this discrepancy.

Table 5. Maximum absolute change in horizon-8 native MSE on the fixed eight-root numerical sample. These are arithmetic/layout contrasts, not physical accuracy. Convolution IEEE is a controlled precision permission, not a new selected model.

Comparison	LeWM	DINO
Observed paths	4.90e-03	1.53e-03
Encoding only (single)	4.89e-03	1.52e-03
Rollout layout (new encoding)	2.56e-06	9.54e-06
Wrapper vs. public	0	0
Metric reduction only	0	1.19e-07
Observed paths, conv. IEEE	2.56e-06	6.44e-06

The numerical audit fixes the first eight original manifest entries and crosses the observed encoding and rollout layouts, retaining actual wrapper and public parity calls. Each model produces 7,680 comparison rows. Unchanged same-process repeats are exact. Encoding differences dominate baseline observed-path changes; rollout-layout effects are much smaller (Table 5). Public and feedback-wrapper outputs agree exactly. Metric reduction alone is zero for LeWM and near float32 rounding scale for DINO.

Changing convolution precision permission from TF32 to IEEE reduces maximum observed-path MSE disagreement by about 1,910-fold/238-fold. Matmul IEEE and cuDNN deterministic controls produce no observed change; explicit matmul TF32 does. Benchmark-off is already set and is explicitly skipped rather than presented as an executed intervention. Parameters, all buffers including nonpersistent masks, and restored backend properties are checked. This establishes controlled numerical sensitivity on the bounded sample, not the cause of every historical maximum or improved physical prediction. Both forecast and encoded reference tensors can shift.

6. Discussion and Limitations

What survives the original hypothesis. The physical editor-transfer claim remains unsupported because its measuring instrument and edited feature surface were not qualified. The independent positive findings concern latent action correspondence. Their coexistence with poor physical readouts is an empirical observation, not a theorem that latent fidelity precludes or guarantees physical understanding. A qualified history- based readout could change the interpretation without invalidating the archived negative result for the original single-image probes.

Scope of decision evidence. We neither prove a causal decomposition of ranking error nor demonstrate closed-loop control. Actual-endpoint ranking is a privileged diagnostic, not a deployable policy, and the limited bank may fail to offer a useful action. The fresh DINO ranking interval is inconclusive. Broader reachable-bank or closed-loop gains require a new design and cannot be inferred from latent effect fidelity alone.

Generalization and multiplicity. There is one environment, two specific unmatched releases, one generated bank distribution and a recorded pretraining-distribution sample. We cannot attribute differences causally to architecture or rank models using their raw latent errors. Follow-ups reuse inspected discovery data; only the fixed five fresh-root contrasts receive primary-family inference. The 99% bootstrap intervals offer nominal/asymptotic multiplicity control, not guaranteed finite-sample coverage. All other results, including fresh feedback and appearance/horizon curves, are exploratory. The bounded probe search does not reproduce every original paper's probing protocol.

Numerical and verification boundary. Same-process repeatability does not establish portability across devices, caches, drivers or historical concurrent workloads. Raw latent tensors were not archived. Independent verification recomputes every CSV summary, candidate- level algebra, ranking choice, bootstrap interval and recorded physical/coverage quantity from saved data; it cannot regenerate all latent measurements without rerunning the models. Hash integrity is not a substitute for scientific validity.

7. Conclusion

On two frozen PushT releases, action-effect fidelity and matched-action specificity replicate on a fixed independent sample, while physical-edit interpretation remains unqualified and a discovery ranking advantage does not receive independent primary-family support. A bounded backend audit further shows that numerical layout must be controlled even with immutable checkpoints. The resulting case study argues for reporting the precise level of evidence earned by an experiment, including failed prerequisites and inconclusive decisions, rather than turning predictive success into broader physical or control claims.

Impact Statement

This work concerns evaluation methodology for simulated manipulation and frozen world models. Its intended benefit is more careful separation of predictive, physical and decision claims. Misinterpreting small latent errors or finite-bank regret as safe physical control could encourage unreliable deployment; our results do not justify deployment on real robots. We use released model and simulation artifacts, not human-subject data. Redistribution of checkpoints, datasets and upstream code remains subject to their respective terms. Scientific and attribution checks, author approval, applicable AI-assistance disclosure and the actual target venue's requirements are necessary before submis-

sion.

References

Chi, C., Feng, S., Du, Y., Xu, Z., Cousineau, E., Burchfiel, B. C., and Song, S. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023. doi: 10.15607/RSS.2023.XIX.026. URL https://www.roboticsproceedings.org/rss19/p026.html.

Geiger, A., Lu, H., Icard, T., and Potts, C. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/4f5c422f4d49a5a807eda27434231040-Abstract.html.

Grant, S., Han, S. J., Tartaglini, A. R., and Potts, C. Addressing divergent representations from causal interventions on neural networks, 2025. URL https://arxiv.org/abs/2511.04638v5. First submitted 2025-11-06; arXiv version 5 revised 2026-04-22.

Huang, H., Wang, Z., Liu, X., Wang, S., Sun, Y., Yu, Z., Ma, J., Cui, S., and Zhang, C. GeoCo-SAVi: Geometry-consistent slot attention for explicitly editable object representations, 2026. URL https://arxiv.org/abs/2609.06628v3. arXiv version 3, revised 2026-09-26.

Joseph, S., Garrido, Q., Balestriero, R., Kowal, M., Fel, T., Bakhtiari, S., Richards, B. A., and Rabbat, M. Interpreting physics in video world models. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 54687–54724. PMLR, 2026. URL https://proceedings.mlr.press/v306/joseph26a.html.

Lambert, N., Amos, B., Yadan, O., and Calandra, R. Objective mismatch in model-based reinforcement learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, volume 120 of *Proceedings of Machine Learning Research*, pp. 761–770. PMLR, 2020. URL https://proceedings.mlr.press/v120/lambert20a.html.

Lippe, P., Magliacane, S., Löwe, S., Asano, Y. M., Cohen, T., and Gavves, S. CITRIS: Causal identifiability from temporal intervened sequences. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 13557–13603. PMLR, 2022. URL https://proceedings.mlr.press/v162/lippe22a.html.

Liu, S., Wu, C., Wen, Q., You, F., Zhang, C., Hao, S., Lin, X., and Su, N. D-JEPA: A decision-aligned latent world model, 2026. URL https://arxiv.org/abs/2609.24749v1. arXiv version 1, submitted 2026-09-21.

Liu, Y. and Chen, Y. Low-rank dynamics-effective latent carriers for counterfactual rollout in learned world models, 2026. URL https://arxiv.org/abs/2608.15156v4. arXiv version 4, revised 2026-09-07.

Maes, L., Le Lidec, Q., Facury, L., Massaudi, N., Chaurasia, A., Capuano, F., Gao, R., Gillin, T., Haramati, D., Scieur, D., LeCun, Y., and Balestriero, R. stable-worldmodel: A platform for reproducible world modeling research and evaluation, 2026a. URL https://arxiv.org/abs/2605.21800v1. arXiv version 1, submitted 2026-05-20.

Maes, L., Le Lidec, Q., Scieur, D., LeCun, Y., and Balestriero, R. LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels, 2026b. URL https://arxiv.org/abs/2603.19312v3. arXiv version 3, revised 2026-06-03.

Oberweger, F. F., Schwingshackl, M., and Murschitz, M. Does latent planning survive point clouds? action-conditioned JEPA world models for geometric observations and goals, 2026. URL https://arxiv.org/abs/2608.29434v2. arXiv version 2, revised 2026-09-28.

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jegou, H., Mairal, J., Labatut, P., Joulin, A., and Bojanowski, P. DINOv2: Learning robust visual features without supervision, 2023. URL https://arxiv.org/abs/2304.07193v2. First submitted 2023-04-14; arXiv version 2 revised 2024-02-02.

Resani, H. and Shi, J. Q. Do-JEPA: From masking to intervention in latent world models, 2026. URL https://arxiv.org/abs/2609.37378v1. arXiv version 1, submitted 2026-09-29.

Subramanian, A. Counterfactual action evaluation, observation bottlenecks, and representation geometry in joint-embedding predictive world models, 2026. URL https://arxiv.org/abs/2610.02860v1. arXiv version 1, submitted 2026-10-02.

Vakalis, D. The intervention gap in latent world models, 2026. URL https://arxiv.org/abs/2608.29998v1. arXiv version 1, submitted 2026-08-30.

Wang, J., Jing, Y., Le, X., Chen, C., Guan, X., and Tao, D. World model interpretability: From decoding representations to understanding latent worlds, 2026. URL https://arxiv.org/abs/2610.25296v1. arXiv version 1, submitted 2026-10-18; perspective article.

Yan, H., Zhu, J., Jia, M., Yin, R., He, J., Zhong, Z., Li, J., Lu, J., Li, H., Zhang, T., Chen, J., Song, W., Chen, W., Gao, Y., and Li, H. Is forward prediction enough? physical state grounding for JEPA world models, 2026. URL https://arxiv.org/abs/2608.06799v1. arXiv version 1, submitted 2026-08-07.

Yao, J., Wang, Z., Huang, Y., and Hu, L. RolloutFaith: Auditing persistent internal interventions in visual world model, 2026. URL https://arxiv.org/abs/2609.36843v1. arXiv version 1, submitted 2026-09-29.

Zhang, Z., Wang, Y., Guan, Z., Yang, Y., Shi, B., Zong, T., Yi, H., Chao, G., Chen, X., Yang, T., Bao, C., Yu, T., Zhou, J., and Xu, J. Delta-JEPA: Learning action-sensitive world models via latent difference decoding, 2026. URL https://arxiv.org/abs/2606.31232v1. arXiv version 1, submitted 2026-06-30.

Zhou, G., Pan, H., LeCun, Y., and Pinto, L. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 79115–79135. PMLR, 2025. URL https://proceedings.mlr.press/v267/zhou25t.html.

A. Artifact Identity and Reproducibility

The experimental record includes exact source/configuration snapshots, fixed rosters and input hashes, selection/plan publication before evaluation, per-stage logs, timestamped status events, bootstrap draws and complete report/data inventories. Output directories are exclusive: no implicit overwrite or resume is used. Numerical failures stop execution; they do not authorize replacement roots. Completed and failed/inconclusive scientific outcomes are different notions. An anonymous release must be prepared before any actual submission; this working PDF intentionally omits identifying project and sandbox links.

Table 6. Released artifact identities. These upstream identities identify the audited checkpoints, not an architecture-wide comparison. SHA256 values are included in the accompanying authenticated evidence ledger.

Component	Upstream source/release	Pinned revision
LeWM source	`lucas-maes/le-wm`	`8edfeb336732b5f3ce7b8b210d0ba370a09e2cac`
LeWM weights	`quentinll/lewm-pusht`	`22b330c28c27ead4bfd1888615af1340e3fe9052`
Point implementation/simulator	`fafraob/point-lewm`	`1b5a90ebf63e8ef165b1dfe100b81b12f4fa1424`
RGB DINO release	`fafraob/point-cloud-pusht`	`664278a4c0b4c5f607d0993744986b2d10b4168e`
DINOv2 source	`facebookresearch/dinov2`	`85a24602099d397264d5b30461ad7f3bfd726ca1`
Recorded PushT data	`quentinll/lewm-pusht`	`655cd446b9929369d7d406001da85c15d1457850`

The original DINO release did not record its hub backbone source revision; the explicit audit pin fixes that ambiguity rather than claiming it was originally recorded. Auxiliary recurrence uses the released embedding, including its nonzero output for zero raw proprioception. Optional xFormers is disabled; native PyTorch attention is used. LeWM and DINO action normalization uses the released population statistics once. Recorded actions are not silently clipped to the native generated-control box.

Hosted inference uses Python 3.11, PyTorch 2.10.0+cu128 and a 97,887-MiB RTX PRO 6000 Blackwell Server Edition. CPU-only contract tests and report/PDF analysis are distinct from research inference. The complete implementation suite at the execution checkpoint contains 1,312 passing tests, no failures or skips. Tests check invariants and real-checkpoint interfaces, not empirical success. The tracked pending batch ran in about 23.34 minutes including numerical controls, input authentication, generation, both model evaluations, analysis and backups. LeWM and DINO fresh model stages took 131.57 and 570.18 seconds, respectively. These are observed wall times, not controlled cross-model speed benchmarks.

Four pending-stage/operations ZIPs were copied off-host and strictly restored, with 610 inventory entries verified in total. Their complete data include 128 fresh pilot and 192 copied development-root archives; repeated source snapshots are counted as inventory entries, not independent data. Independent verifiers recompute numerical summaries from every CSV row and replication intervals from every complete bank. Neither loads checkpoints nor simulates new roots.

B. Probe Selection, Observability and the Initial Failure

Ridge pooling grids are 4 and 7 with regularization $\{0.01, 1, 100, 10000\}$. LeWM's global projected token makes spatial pooling degenerate; it is not assigned DINO's patch geometry. MLP pooling grid is 4, hidden width 64 with SiLU, learning rate 0.001, batch size 128, weight decay $\{0, 0.01\}$ and checkpoints at epochs $\{50, 100, 200\}$. Seeds $\{11, 29, 47\}$ use isolated initialization and shuffle streams. Selections are target-group-specific and based on original-appearance validation roots, with canonical ties and no pilot-based best-seed choice.

The MLP standardizes feature coordinates using scorer means and population standard deviations, replacing deviations at most 10^{-6} by one. Its targets use scorer means and fixed training scales $s_{\mathrm{MLP}} = (512, 512, 512, 512, 1, 1)$. Separately, validation/reporting and the empirical ambiguity calculation use $s_{\mathrm{report}} = (32, 32, 32, 32, 0.25, 0.25)$ on the six XY/sine/cosine labels. The scaled squared loss is a *sum*, not an average over coordinates:

$$L_s(\hat{y}, y) = \sum_{j=1}^{6} \left((\hat{y}_j - y_j)/s_j\right)^2 . \tag{4}$$

Target-group selection uses its group's summands, with equal-root aggregation. Native-distance/angular metrics are reported separately. Thus the draft 0.05 scaled-loss gate and the empirical 3.6240 lower bound are not errors in degrees or arena units, and are not the MLP training-loss scale.

Training/validation use both branches, context/future horizons and candidate index equal to root index modulo eight. Pilot evaluation uses every candidate, both branches and every appearance. At original appearance it contains 6,912 image samples, including 2,048 horizon-8 samples. The prior prerequisite readout used a selected-candidate mixed-time summary of 1,536 images; the two tables do not measure identical sample averages. The original editor pilot also used mixed actual and predicted scoring surfaces. Its reported scaled losses must not be compared numerically as though they were the later native distance MAEs.

For an exact resized RGB observation x with multiple scaled physical labels y, the minimizing deterministic single-image squared-loss predictor is the within-x mean. Summing the within-group variance over this empirical labeled sample gives the reported lower bound 3.6240. That statement concerns exact pixel matches and the defined squared loss only. It is not a population irreducible-error estimate or a lower bound for history input, angular MAE, Euclidean distance or a differently constructed dataset. Geometric off-canvas counts likewise do not certify semantic invisibility or justify excluding samples after observing outcomes.

The unresolved original editor-transfer hypothesis remains in the evidence record. Positive MLP-minus-identity scored effect loss under an unqualified scorer does not establish physically harmful edits. Transport interaction intervals spanning zero neither establish invariance nor an appearance-specific failure mechanism.

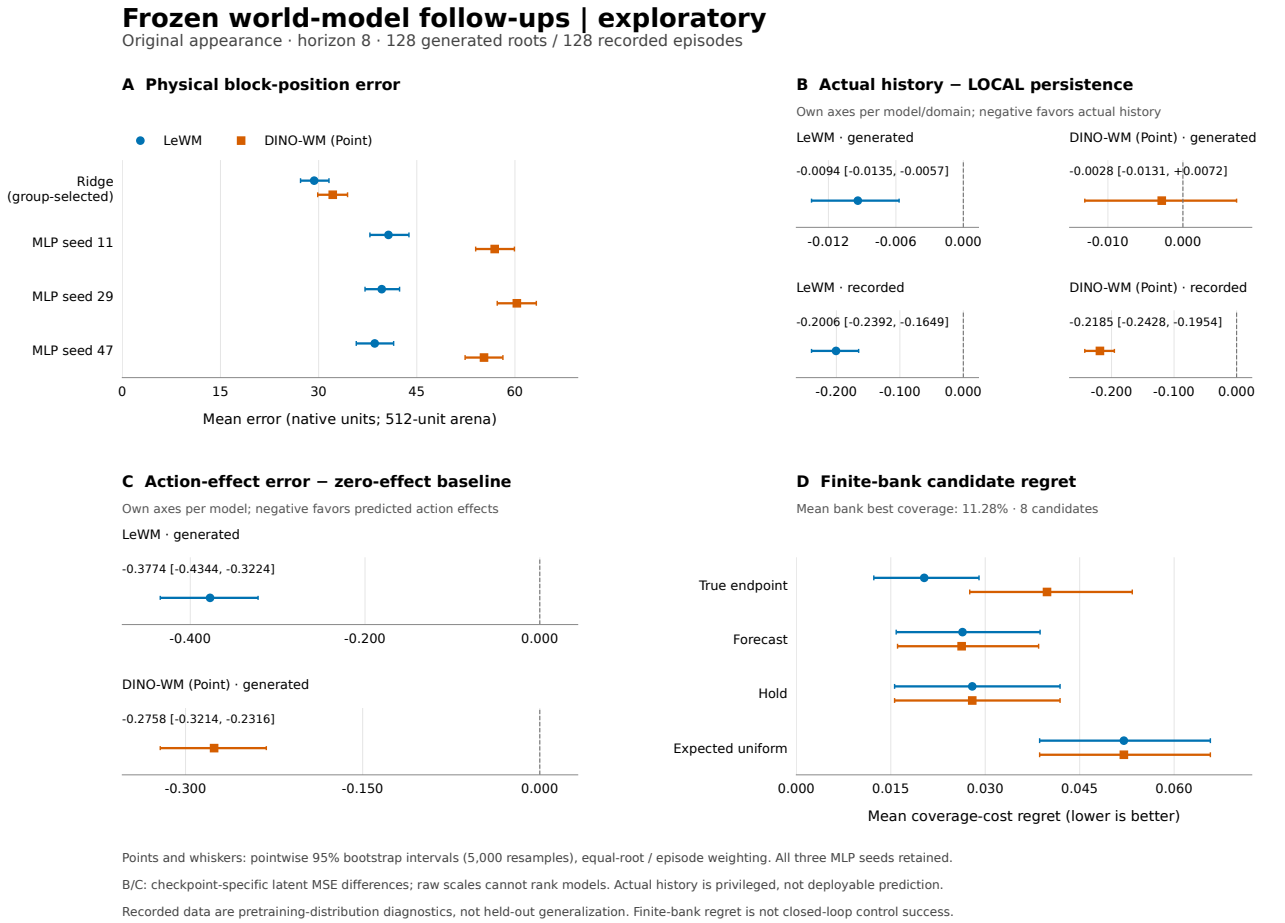


Figure 2. Authenticated discovery follow-up summary. All probe seeds, oracle/ local-persistence controls, action contrasts and finite-bank regrets are exploratory pointwise 95% views on inspected data. These intervals must not be read as the independent primary-family results in Figure 1.

C. Primary and Complementary Inference

The original manifest includes all five split names, including an empty confirmation roster. Every original archive is authenticated before fresh-root generation. Fresh root IDs, 128-bit root seeds and environment seeds are checked against old and reserved identities. The generator retains the native bank, clearance 2 and maximum 256 initial-anchor attempts; no bank improvement, contact/visibility screening or outcome filtering is inserted.

Root means are ordered by full root ID. The saved 10000×128 int64 bootstrap matrix is generated once by NumPy PCG64 with seed 20261102 and used for both models and all contrasts. Primary quantiles are 0.005/0.995, linear interpolation. The five-member family includes Equations 1 and 2 for each model and forecast-minus-endpoint regret for DINO only. Every primary estimate and interval is reported even if another member is inconclusive. All-primary replication requires all five; that criterion was not met.

The effect gain is $g = \overline{\langle \Delta P, \Delta A \rangle} / \overline{\|\Delta A\|^2}$, using equal-root means after candidate and branch averaging. If any resampled denominator is zero, the unconditional interval is undefined and the count is reported rather than conditioning on nonzero draws. Both fresh gains have zero undefined draws. An energy ratio near one and $g < 1$ are compatible with misaligned/off-axis effects; a pure insufficient-amplitude explanation is not established.

The untouched underlying dynamics reports retain their legacy exploratory schema and inspected-pilot boilerplate. Fresh sample identity and primary inference are established by the new plan, composite data manifest and joint analysis, not by rewriting legacy report tags. All 1,680 exploratory estimates/intervals and five primary contrasts were independently recomputed; maximum comparison discrepancy was 2.27×10^{-13}. The numerical verifier checks 1,800 summary statistics per model with rounding discrepancy at most 1.11×10^{-16}.

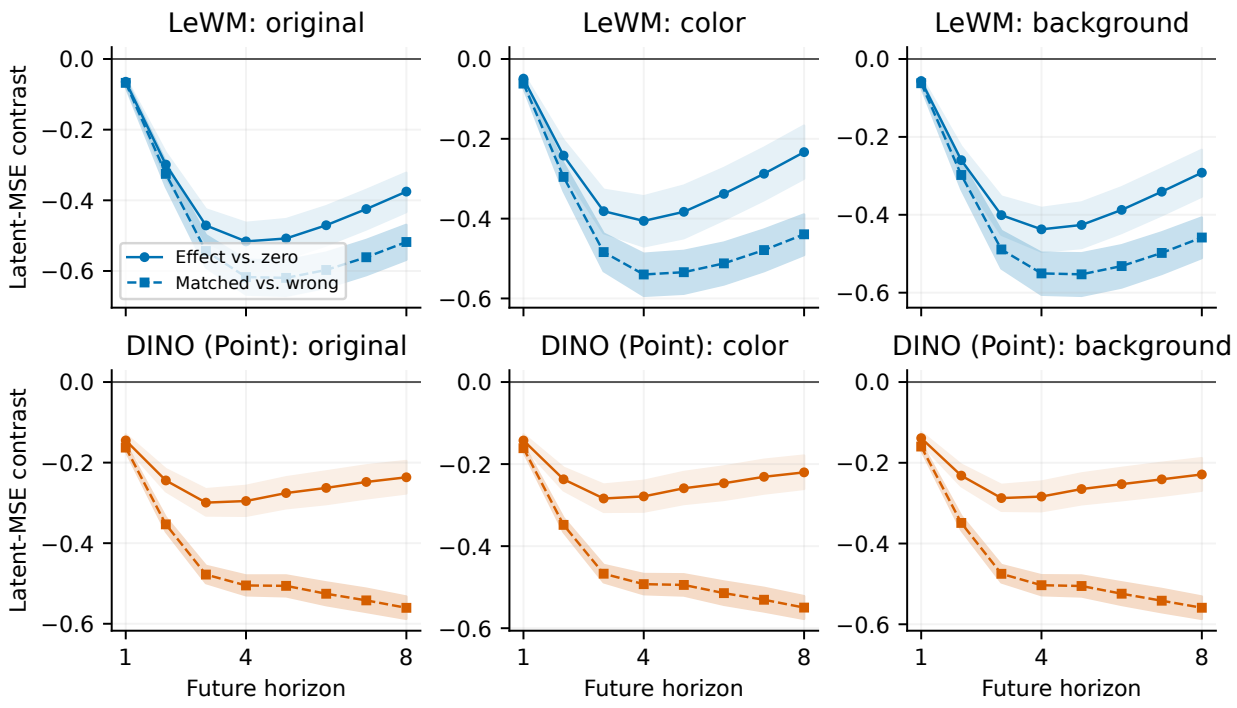


Figure 3. All fresh-sample horizons and appearances for the two action contrasts. Shaded intervals are pointwise exploratory 95%, including the primary metrics' exploratory views. Only original appearance/horizon 8 uses the dedicated 99% primary family. No favorable slice is selected to replace the inconclusive decision result. Raw latent scales remain model-specific.

D. Numerical Audit Matrix and Limits

The eight-root numerical audit retains both branches, all candidates and common actual horizons 1/4/8. The old path encodes context as batches 4,2, encodes each candidate's three selected future frames together, and invokes public single-candidate native and zero-action rollouts. The new path retains context/goal/future encoding calls, the actual free/oracle control suite

Table 7. Complementary fresh-sample summaries at every appearance and the original reporting horizons. Every interval is pointwise exploratory 95%. Fidelity and feedback use model-specific latent MSE; forecast–hold uses native coverage-cost regret. No row is an additional primary claim.

Model	Appearance	h	Oracle–local persistence	Effect–zero	Forecast–hold
LeWM	original	1	-0.0529 [-0.0633, -0.0435]	-0.0645 [-0.0762, -0.0539]	-0.0008 [-0.0038, 0.0022]
LeWM	original	4	-0.0572 [-0.0693, -0.0463]	-0.5162 [-0.5715, -0.4639]	0.0063 [-0.0001, 0.0133]
LeWM	original	8	-0.0112 [-0.0148, -0.0078]	-0.3753 [-0.4329, -0.3221]	0.0034 [-0.0047, 0.0116]
LeWM	color	1	-0.0389 [-0.0501, -0.0285]	-0.0493 [-0.0618, -0.0377]	-0.0010 [-0.0038, 0.0018]
LeWM	color	4	-0.0430 [-0.0559, -0.0312]	-0.4057 [-0.4695, -0.3441]	0.0073 [-0.0012, 0.0164]
LeWM	color	8	0.0038 [-0.0018, 0.0098]	-0.2333 [-0.2996, -0.1671]	0.0051 [-0.0019, 0.0123]
LeWM	background	1	-0.0459 [-0.0568, -0.0363]	-0.0566 [-0.0689, -0.0457]	-0.0002 [-0.0033, 0.0028]
LeWM	background	4	-0.0532 [-0.0654, -0.0420]	-0.4374 [-0.4943, -0.3825]	0.0076 [-0.0006, 0.0163]
LeWM	background	8	-0.0069 [-0.0109, -0.0032]	-0.2920 [-0.3531, -0.2334]	0.0085 [0.0001, 0.0178]
DINO (Point)	original	1	-0.0625 [-0.0784, -0.0459]	-0.1450 [-0.1592, -0.1301]	-0.0016 [-0.0042, 0.0004]
DINO (Point)	original	4	-0.0015 [-0.0151, 0.0116]	-0.2953 [-0.3319, -0.2578]	-0.0079 [-0.0154, -0.0018]
DINO (Point)	original	8	0.0029 [-0.0070, 0.0130]	-0.2364 [-0.2763, -0.1961]	-0.0033 [-0.0127, 0.0046]
DINO (Point)	color	1	-0.0575 [-0.0738, -0.0406]	-0.1425 [-0.1567, -0.1276]	-0.0007 [-0.0032, 0.0014]
DINO (Point)	color	4	0.0049 [-0.0081, 0.0175]	-0.2787 [-0.3158, -0.2409]	-0.0080 [-0.0155, -0.0020]
DINO (Point)	color	8	0.0036 [-0.0060, 0.0135]	-0.2202 [-0.2603, -0.1793]	-0.0040 [-0.0134, 0.0040]
DINO (Point)	background	1	-0.0317 [-0.0479, -0.0148]	-0.1389 [-0.1527, -0.1245]	-0.0012 [-0.0033, 0.0005]
DINO (Point)	background	4	0.0323 [0.0186, 0.0459]	-0.2837 [-0.3200, -0.2467]	-0.0068 [-0.0140, -0.0011]
DINO (Point)	background	8	0.0330 [0.0226, 0.0436]	-0.2289 [-0.2691, -0.1886]	-0.0054 [-0.0155, 0.0030]

and its public eight-candidate parity call. Controlled bridges cross old/new encoded context and targets with single/bank public rollouts. The bank layout includes observed strides; it does not isolate GEMM batch size alone.

Each observed suite executes twice with unchanged incoming backend settings. Four distinct controls are then applied individually: matmul IEEE, matmul TF32, convolution IEEE and cuDNN deterministic. Benchmark-off is already off. Every property is restored in a finally block, including precision inheritance; unavailable legacy getters under modern overrides are recorded explicitly. This yields seven within-setting comparisons for baseline and each control, five repeat comparisons and five baseline/control variant comparisons per control: 60 groups with 128 matched records each. The controls were specified before inspecting their outcomes.

Native float32 and common CPU float64 metric reductions are stored. Tensor maximum/mean absolute differences have full root/branch/candidate/horizon identities. Same-layout repeats, matmul IEEE and cuDNN deterministic produce no observed changes on this sample. Other settings shift the encoded targets as well as predictions. Permissions do not prove which backend kernel was selected. CUDA/cuDNN cache history, driver, attention backend, allocator and concurrent workloads are not reset or exhaustively varied. Historical backend state and full raw tensors were not saved, so historical causal attribution and cross-process portability remain unresolved.

E. Reporting and Extension Boundaries

This paper is an evidence checkpoint, not a commitment to obtain a positive result by further search. Future observability-aware history probes, state-specific action-effect readouts and contact/reachable-bank or closed-loop decisions would need separate hypotheses, competing alternatives, frozen selection, fresh evaluation and failure-preserving protocols. They cannot be inserted as results until actually run and verified. A qualified editor study depends first on physical-change measurement at adequate resolution. Any new bank changes the task distribution and must be labeled accordingly.

No claims of priority for diagnostic separation or a new world-model method are made. Close grounding, intervention and decision-alignment papers are cited in Section 2. Final bibliographic versions, venue rules, anonymity, author contributions, disclosure and redistribution terms require author review. ICML style is a typesetting choice for this draft, not evidence of submission, acceptance or compliance with a future call.